

Testing & Validation Record

HealthConnect Clinic Experience Lab — Week 7 Analytics QA

PROJECT TRACK	PHASE / MILESTONE	LEAD ANALYST	OVERALL STATUS
Data Analytics	Week 7 Quality Assurance	Analytics Team	PASSED (Internal)

1. OBJECTIVE & APPROACH

In Week 7, we shifted focus from broad exploratory analysis to rigorous testing, validation, and refinement. The main goal was to make sure that all numbers produced during Week 6 are accurate, repeatable, and aligned across all project outputs (Python notebook, Power BI dashboard, and written findings).

Rather than introducing new analytical methods, this phase verified that our data pipeline is solid, our dashboard visuals respond accurately to filters, and our business recommendations directly reflect the underlying data.

2. KPI & DATA INTEGRITY VALIDATION

We re-ran the full dataset in Python to verify the core metrics. The baseline dataset contains exactly **4,737 eligible appointments**. We verified that attendance and no-show counts sum up to the total, and that the rates total 100%.

```
=== PYTHON BACKEND VALIDATION OUTPUT === Eligible appointments: 4737 Attended: 2314
No-show: 2423 Attendance rate: 48.85 % No-show rate: 51.15 % Counts add up: True Rates add
up: 100.0 % Priority Segment (Prev No-Show + 31-60d Lead): 977 appts | 70.52% No-Show
```

Metric / KPI	Python Output	Power BI Card	Reconciliation	Result
Eligible Appointments	4,737	5K (Rounded)	Exact match in data model	PASS
Attended Appointments	2,314	2.3K (Implicit)	2,314 / 4,737 = 48.85%	PASS
Total No-Shows	2,423	2K (Rounded)	Exact match in data model	PASS
Attendance Rate	48.85%	0.49	Formatted as 2-dec decimal	PASS
No-Show Rate	51.15%	0.51	Formatted as 2-dec decimal	PASS

3. PRIORITY RISK SEGMENT VERIFICATION

We validated the primary high-risk group identified in Week 6: patients with a **previous no-show history booking 31 to 60 days in advance**. Across 977 appointments fitting this profile, the no-show rate reached **70.52%**. This verified pattern serves as the foundation for our targeted intervention strategy.

4. POWER BI INTERACTIVE DASHBOARD SLICER TESTING

To test the interactive integrity of the Power BI dashboard (HealthConnect_Week6.pbix), we applied specific visual slicers and cross-checked the updated cards, charts, and matrix tables against sub-queries.

Slicer Applied	Filter Selection	Observed Dashboard Behavior	Status
Appointment Type	Follow-up	Cards updated to 1K Eligible, 728 No-Shows, 0.46 Attendance, 0.54 No-Show. Visuals filtered correctly.	PASS
Appointment Type	Diagnostic Test	Cards updated to 567 Eligible, 295 No-Shows, 0.48 Attendance, 0.52 No-Show. Visuals filtered correctly.	PASS
Heatmap Matrix	Lead Time x Prior No-Shows	Conditional blue-to-red color gradient dynamically shifts based on cell value intensity across all slicer states.	PASS

5. RECOMMENDATION TRACEABILITY MATRIX

Every recommendation was evaluated to ensure direct alignment with validated analytical evidence, avoiding unsubstantiated assumptions or false causal claims.

Validated Finding	Derived Practical Recommendation	Traceability Status
Overall No-Show rate (51.15%) exceeds Attendance rate (48.85%).	Establish automated attendance tracking and maintain continuous KPI monitoring.	VERIFIED
Peak no-show rates occur in the 31–60 day lead time window.	Schedule secondary reminder notifications at Day 14 and Day 3 prior to long-lead appointments.	VERIFIED
Prior no-show patients with >30 days lead time reach a 70.52% no-show rate.	Treat this high-risk combination as a priority segment for confirmation calls or double-booking buffers.	VERIFIED
Reminder channel logs contain missing values.	Refrain from claiming channel causality until tracking data quality is improved.	VERIFIED

6. LIMITATIONS & CROSS-TRACK HANDOFF STATUS

Cross-Track Data Science Collaboration: PENDING

A formal handoff dataset was structured during Week 6 to support predictive machine learning models. However, joint cross-track validation with the Data Science team was not formally executed during Week 7. The findings in this document represent complete **Data Analytics track validation**, while predictive modeling validation remains open for future integration cycles.

7. CONCLUSION & SIGN-OFF

The Week 7 testing cycle confirms that the HealthConnect analytical model, Python metrics, and Power BI dashboard